

$$\textcircled{1} \quad y = \underline{x^3 + ce^{-3x}} \quad \dots (1)$$

$$y' = \underline{3x^2 - 3ce^{-3x}} \quad \dots (2)$$

$$y = x^3 + \frac{3x^2 - y'}{3} \rightarrow$$

$$3y = \underline{x^3 + 3x^2 - y'} \Rightarrow y' + 3y - 3x^2(x+1) = 0$$

$$\Rightarrow y' + 3y - 3x^2(x+1) = 0. \quad \underline{\underline{DE}}$$

$$\textcircled{2} \quad y = \underline{c_1 e^{-x}} + \underline{c_2 e^{2x}} \quad \checkmark$$

$$y' = -c_1 e^{-x} + 2c_2 e^{2x}$$

$$y'' = c_1 e^{-x} + 4c_2 e^{2x}$$

$$y + y' = 3c_2 e^{2x}$$

$$y' + y'' = \underline{6c_2 e^{2x}}$$

$$y' + y'' = 2(y + y')$$

$$\Rightarrow y'' - y' - 2y = 0$$

↑

$$\textcircled{3} \quad y = \underline{c_1 \sin 2x + c_2 \cos 2x}, \quad \text{DE: } \underline{\underline{y'' + 4y = 0}}$$

$$y' = 2c_1 \cos 2x - 2c_2 \sin 2x$$

$$y'' = \underline{-4c_1 \sin 2x} - \underline{4c_2 \cos 2x}$$

$$y'' = -4xy \Rightarrow \underline{\underline{y'' + 4y = 0}}$$

$$\underline{y'' + 3y' + 2y = \underline{3e^{5x}}.} \quad g(x) = 3e^{5x}.$$

y_c is the solution to $y'' + 3y' + 2y = 0$

$$\lambda^2 + 3\lambda + 2 = 0$$

$$\lambda = -1, -2$$

$$y_c = c_1 e^{-x} + c_2 e^{-2x}$$

$$y_p = \underline{c e^{5x}}, \quad y_p' = 5 \underline{c e^{5x}}, \quad y_p'' = 25 \underline{c e^{5x}}.$$

$$\cancel{c e^{5x}} (25 + 3 \times 5 + 2) = 3 \cancel{e^{5x}}$$

$$\Rightarrow c \times 42 = 3$$

$$\Rightarrow c = 3/42$$

$$y_p = \frac{3}{42} e^{5x}.$$

$$y = c_1 e^{-x} + c_2 e^{-2x} + \frac{3}{42} e^{5x}.$$

$$b) \underline{y'' + 3y' + 2y} = \underline{x^2}$$

$$y_c = c_1 e^{-x} + c_2 e^{-2x}$$

$$\text{Try } y_p = p_0 + p_1 x + p_2 x^2, \quad y_p' = p_1 + 2p_2 x, \quad y_p'' = 2p_2$$

$$2p_2 + 3(p_1 + 2p_2 x) + 2(p_0 + p_1 x + p_2 x^2) = x^2$$

$$\underline{2p_2 = 1}, \quad \underset{\downarrow}{6p_2 + 2p_1 = 0}, \quad \underset{\downarrow}{2p_2} + \underset{\downarrow}{3p_1} + \underset{\downarrow}{2p_0} = 0$$

$$\underline{p_0 = 7/4, \quad p_1 = -3/2, \quad p_2 = 1/2}$$